

Identifying Design Opportunities for Behavior Displacement for Intervening Sedentary Time among Older Adults

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Prolonged sedentary behavior poses significant health risks, calling for interventions that promote active lifestyles. For older adults, every physical activity, no matter how small or significant, plays a vital role in their quality of life. However, many interventions aimed at reducing sedentary behavior have overlooked the unique needs and preferences of older adults. In this study, we explore design opportunities for supporting behavior displacement—replacing sedentary time with active movements—as a potential strategy for intervening sedentary time among older adults. Through a 7-day diary study and interviews with 13 participants, we uncovered key factors, such as attention demand, productivity and quality of activities, physical fatigue, as well as social norms, that influence their decisions to engage in displacement. We also identified sequential and concurrent displacement strategies and the contexts in which each was employed. Our findings highlight the need for designing personalized, adaptive interventions that respect the diverse preferences and agency of older adults.

CCS Concepts: • **Human-centered computing** → **Empirical studies in HCI**.

Additional Key Words and Phrases: Sedentary behavior; older adults; physical activity; behavior displacement; diary study; interview

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1 INTRODUCTION

The high prevalence of sedentary behavior, characterized by long periods of minimal physical activity and extensive screen time [64], has become a significant concern. Studies show that older adults spend between 65 to 80% of their waking hours engaged in sedentary activities [89]. This behavior is linked to a variety of negative health outcomes particularly for older adults, such as higher risks of sarcopenic obesity [69], lower bone mineral density [60], and increased cardio-metabolic risks [31]. Recognizing these health risks among older adults has led to the development of interventions aimed at promoting more active lifestyles, incorporating traditional recommendations such as 150–300 minutes/week of moderate to vigorous physical activity (MVPA) [66]. However, such recommendation might be too demanding for older adults, who often face challenges such as reduced mobility, chronic health conditions [23], and perceived barriers to the feasibility of high-intensity activities [90]. These issues often hinder their ability to sustain the recommended levels of physical activity, contributing to high dropout rates from exercise programs [47].

Evidence suggests that even small amounts of physical activity, such as indoor walking and stair climbing, can offer health benefits, particularly for older adults, highlighting the importance of reducing sedentary behavior in any way possible, as every bit of movement counts toward improving overall health [83].

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An effective approach to promote physical activity in older adults is to replace sedentary behavior with more active movements. This strategy, known as *displacement*, involves replacing sedentary behaviors with alternatives that are less sedentary or those that require greater movement [71].

While traditional substitution models primarily emphasize the long-term health effects of replacing one activity with another in a way that keeps the total time constant [78], displacement strategies focus more on immediate and practical applications that integrate physical activity into existing routines or real-world settings, making them more achievable in daily contexts. For example, standing or walking meetings [19], taking short active breaks during prolonged sitting periods [24], and using standing desks [65] are practical displacement strategies that can effectively reduce sedentary time without altering task routines or increasing structured exercise sessions. Although these approaches have shown promise across various groups, particularly among younger working individuals, more research is needed to adapt and tailor them for older adults. Studies suggest that simple changes to the physical environment [35], along with self-monitoring [16] and problem-solving [48] can support individuals in integrating displacement activities into their daily routines. However, the extent to which older adults can and choose to adopt these behavior displacement strategies in various contexts remains underexplored. Understanding how technology can support behavior displacement among older adults is an important and promising topic for UbiComp community, offering valuable opportunities to examine user needs and preferences, and advance the use of context-aware techniques. Further research is needed to investigate the specific contexts of older adults' sedentary behaviors, their willingness to engage in displacement strategies, their preferences in integrating them into their lifestyle, and the impact of these strategies on the productivity of the displaced sedentary activities.

In light of this, we aim to explore the decision-making processes and strategies older adults use to displace sedentary behaviors. By examining the contexts and preferences for behavior displacement, we seek to understand how such displacement strategies can be effectively integrated into the daily routines of older adults, particularly those whose lifestyles involve sedentary behaviors. Specifically, we want to address the following research question: **How do older adults decide whether and how to displace their sedentary behavior?** We conducted a pre-study questionnaire to capture participants' self-reported sedentary and active levels, followed by a 7-day diary study where participants documented their daily sedentary activities and thoughts on displacement. Finally, we conducted debriefing interviews to further explore participants' routines and reasons for displacing or not displacing certain activities.

Through a codebook thematic analysis [8] of the diary and interview data, we gained insights into the participants' experiences and preferences, identifying the factors influencing their willingness to adopt displacement strategies. Our findings revealed how older adults determine whether to displace sedentary behaviors by considering factors such as attention demand, productivity, physical fatigue, and social norms. We also showed how older adults implement displacement through sequential and concurrent strategies that align with their routines. Our participants were not passive recipients of health recommendations but active agents who carefully weighed the benefits and drawbacks of sedentary activities and displacement strategies. The key contributions of this work are the following:

- We present key factors influencing older adults' decision-making processes for displacing their sedentary behavior, as identified through our 7-day diary study and interviews.
- We propose a framework that categorizes displacement strategies and connects them to behavioral goals, illustrating how older adults integrate displacement strategies into their daily routines.
- Our findings offer insights into how technological interventions can be designed to be more adaptive and context-aware, respecting older adults' diverse routines and preferences. By considering both health and personal values in intervention design, we propose a shift from generic activity tracking systems to more personalized and agency-driven technologies.

2 RELATED WORK

In this section, we first provide an overview of sedentary behaviors among older adults, emphasizing their health risks and the complexity of their social and cognitive benefits. We then review prior research on displacement and intervention strategies targeted at sedentary time. Finally, we discuss the role of self-tracking technologies in promoting behavior change and highlight gaps in addressing the diverse contexts and agency of older adults.

2.1 Older Adults' Sedentary Activities

Sedentary behaviors, defined as waking activities involving low energy expenditure while in a sitting or reclining posture [39, 61], have been extensively studied for their health implications across all age groups [22, 80]. Among older adults, evidence has revealed the risks associated with prolonged sedentary behavior including metabolic syndrome [70] and depression [84]. Studies suggest that older adults may tend to spend 60-70% of their waking hours in uninterrupted bouts of sedentary time, such as reading, knitting, or attending social gatherings [75]. Screen-based activities, such as watching television or using digital devices like smartphones and tablets, also form a significant subset of sedentary routines among older adults [2].

Considering the diversity and heterogeneity of the aging population, it is crucial to acknowledge that this demographic's engagement in sedentary activities is multifaceted, reflecting a blend of personal choice, environmental constraints, and behavioral factors in sedentary time accumulation [12]. Studies emphasized the diverse determinants of sedentary behavior and the varying perceptions of sedentary activities' value among older adults [46]. This variability influences the effectiveness of interventions designed to encourage displacement with physical activities, suggesting a need for tailored approaches that respect the unique preferences and circumstances of older adults.

Despite the health risks, some sedentary activities hold meaningfulness for older adults, offering opportunities for social interaction [51], cognitive engagement [79], and relaxation [62]. For instance, Compennolle et al. synthesized qualitative findings showing that while some older adults recognize the health risks, others value the social and cognitive benefits of activities like reading, watching TV, and engaging in hobbies [15]. This duality underscores the complexity of designing interventions to reduce sedentary time among older adults. Further, Wang et al. revealed that older adults' preferences for activity tracking were influenced by perceived meaningfulness of both active and sedentary activities, arguing for personalized technology design that respects older adults' diverse values [87]. Additionally, Leask et al. highlighted how daily patterns and contexts can inform more effective intervention timing, particularly emphasizing the home environment and afternoon hours as critical targets for reducing sedentary time [41].

Moreover, older adults often exhibit positive attitudes and perceptions towards certain sedentary behaviors, reflecting a complex and multifaceted understanding of sedentary lifestyles. Studies suggest that older adults can engage in sedentary behavior due to habits and lack of awareness, pointing to the interventions that increase awareness and promote active alternatives [45]. Ramalho and Petrica further highlight that some sedentary behaviors, when active and socially engaging, can have protective effects on mental health and cognitive function, suggesting that not all sedentary activities are equally detrimental [68].

This landscape of research on sedentary activities among older adults reveals a complex interplay of factors influencing sedentary behavior. While considerable progress has been made in understanding the scope and impact of sedentary behavior among older adults, gaps remain in capturing more finer details of their daily sedentary routines, and how these factors can affect older adults' decision-making on displacing their activities. Through a 7-day diary study, we seek to address these gaps by exploring the specific contexts, purposes, and perceptions of displacing sedentary activities, paving the way for more targeted and effective interventions to promote active living among older adults.

2.2 Displacement and Sedentary Time

Displacement, within the context of sedentary behavior and physical activity, refers to the reallocation of time from one activity to another [28, 52], a concept central to the isotemporal substitution model in epidemiology. In this model, “substitution” specifically involves replacing one type of activity (sedentary) with another (physical activity) to assess potential health impacts [52], with a narrower focus on measurable health benefits [26]. Research in sports science also emphasizes the health enhancements associated with replacing sedentary activities with physical engagements [55, 78]. In contrast, our study adopts “displacement” in line with psychological and behavioral perspectives, referring more broadly to “reallocation”. Rhodes and Blanchard explored “displacement” as the systematic redistribution of time among different activities [71]. They revealed that limited leisure time decreased individuals’ confidence in displacing preferred sedentary behaviors with physical activity, highlighting the impact of psychological factors on the displacement of sedentary time [71]. This angle complements the broader discussion on displacement by identifying motivational and contextual constraints as critical factors in the displacement of sedentary time [27, 72].

Research further supports the value of displacement strategies in improving health outcomes. Healy et al. suggested that even short breaks in sedentary time can offer benefits to metabolic health [33]. Similarly, Grieco et al. showed that integrating physical activity into traditionally sedentary settings, like classrooms, not only enhances students’ engagement but effectively displaces sedentary time [29]. In workplace contexts, strategies such as standing or walking meetings, as well as the use of sit-to-stand desks have been effective in promoting active behaviors without necessarily increasing structured exercise sessions [65, 74]. While these interventions have been widely studied in general populations, specific work focusing on displacement in sedentary time among older adults is less prevalent, despite their unique needs and barriers [17].

Prior Human-Computer Interaction (HCI) research on displacement interventions has explored technologies such as wearable activity trackers and web-based tracking tools for real-time feedback [40, 53], gamified elements to enhance motivation [1, 34], and personalized goal-setting and progress tracking [3, 38]. Other efforts have demonstrated the feasibility of passive, device-free systems using ambient infrastructure, such as WiFi-based sensing, to support personalized fitness tracking in everyday contexts [30, 86]. These technologies demonstrate the potential to encourage behavior displacement by making physical activity more engaging and accessible. However, while earlier studies highlight health benefits, they often lack a unified framework to categorize strategies and connect them to behavior goals [25]. This gap results in an absence of a clear structure for understanding how different displacement strategies are implemented in individuals’ daily routines.

The principles of behavior displacement—such as encouraging short active breaks during prolonged sitting periods [50]—remain applicable and beneficial for older adults. Generally, previous interventions have often emphasized enhancing awareness about health risks of prolonged sedentary behaviors and suggesting physical activity as a feasible substitute [5, 6]. However, such interventions have tended to overlook older adults’ agency—their capacity to make informed decisions and exercise control over their activities, which is essential for older adults’ autonomy, independence and well-being [37]. Recent studies on just-in-time interventions further showed that poorly timed, repetitive, or misaligned prompts can lead to disengagement, highlighting the need for user-centered, context-aware systems [63]. Recognizing and supporting agency is crucial, as it empowers older adults to engage more meaningfully in displacement, thereby enhancing the effectiveness of interventions.

Recognizing this oversight, we explore the opportunities for displacement by analyzing empirical data from diary and interview studies with older adults, which provide the contexts, barriers, and facilitators associated with displacement strategies in their daily lives. Acknowledging the complexity of sedentary behaviors, our work aims to offer insights on how to integrate displacement into the sedentary routines of older adults in a way that enhances the productivity, enjoyment, and alignment of their activities with personal values.

Table 1. Participants' demographics, such as age, gender, employment status, latest occupation, and education, and the self-reported average daily activity in the pre-study questionnaire, including the time spent on moderate-to-vigorous activity (MVPA, at least 10m at a time), the time spent on walking (at least 10m at a time), and the time spent sedentary (sitting and lying while waking)

ID	Age (Gender)	Employment	Latest Occupation	Education	Self-reported Activity/day		
					MVPA	Walking	Sedentary
P1	66 (F)	Fully retired	Addiction counselor	Master's	10m	1h 30m	6h
P2	73 (F)	Part-time work	Educational therapist	Master's	1h	1h	6h
P3	69 (F)	Part-time work	Chemistry teacher	Master's	0	30m	10h
P4	74 (F)	Fully retired	Claim adjuster	High school	20m	40m	6h
P5	83 (F)	Part-time work	Nonprofit organizational consultant	Ph.D./M.D.	1h	1h	6h
P6	77 (F)	Fully retired	Rehabilitation counselor	Master's	0	3h	8h
P7	65 (M)	Part-time work	Healthcare management	Master's	20m	1h	11h
P8	65 (M)	Part-time work	Peer navigator	Bachelor's	1h	30m	7h
P9	73 (F)	Part-time work	Psychotherapist	Master's	1h	1h 30m	9h
P10	67 (M)	Fully retired	Teacher	Master's	50m	40m	6h
P11	70 (F)	Fully retired	Focus group researcher	Master's	15m	1h	8h
P12	82 (M)	Fully retired	Professor	Ph.D./M.D.	30m	2h	7h
P13	70 (F)	Fully retired	Teacher	Bachelor's	1h 30m	2h	7h

3 METHODS

3.1 Participants

We recruited 13 older adults (9 females and 4 males, as detailed in Table 1) from local senior community mailing lists and snowball sampling in the US, aged between 65 and 83 ($M = 71.8$, $SD = 6.0$). This sample exhibited diverse professional backgrounds and employment status, with 7 fully retired and 6 engaged in part-time or voluntary work. Participants were required to (a) live in the United States, (b) be 65 years old or above, (c) retire from their full-time occupation, and might be currently involved in either voluntary or unpaid work or hold a part-time job, but not exceed four hours of sedentary work-related activity per day, (d) spend at least six hours on sedentary behavior per day during waking hours, (e) not have significant mobility issues that might prevent them from engaging in basic physical activities. They also required a stable Wi-Fi connection at home, devices to join video calls, and proficiency in English. Participants were asked to self-report their age, sedentary level, and health conditions in the screening questionnaire, which was included in the recruitment advertisement or sent to them after they contacted the researcher.

3.2 Study Design

The study protocol included a pre-study questionnaire, tutorial session, a 7-day diary study, and a debriefing interview. Besides one participant who opted for an in-person tutorial session, all other participants engaged in remote procedures via Zoom. The study was approved by the authors' University Institutional Review Board, and all procedures were conducted between January and February 2024.

3.2.1 Pre-study Questionnaire. Once participants gave consent to participate in the study, they completed a pre-study questionnaire that included questions modeled after the International Physical Activity Questionnaire

(IPAQ), enabling participants to self-report their activity levels [18, 42]. We asked participants to reflect on and report on the frequency of their sedentary behaviors, such as time spent sitting or lying down with no apparent breaks, participation in vigorous and moderate physical activity and walking over the past week, as well as total weekday sitting time (see Table 1).

3.2.2 Tutorial and 7-day Diary Study. The goal of the 7-day diary study was to capture detailed accounts of older adults' sedentary activities, including the sedentary postures adopted during both non-screen-based and screen-based activities in participants' routines, as well as their willingness to intervene or not. Before the diary study, participants completed a 30-minute one-on-one tutorial. In this session, we explained the study's purpose—to explore how older adults decide whether to displace their sedentary behavior and what displacement strategies they adopt when doing so—and introduced participants to the concept of displacement along with common example scenarios. Participants were encouraged to reflect on their willingness to displace their sedentary behaviors in the diary entries, including any cues, strategies, or barriers they experienced. To ensure their understanding, participants practiced completing the diary using a daily activity as an example. Participants completed the diary study using either an online Google Form or a physical journal. The diary was designed to capture specific instances of sedentary activities, including details on the type of activity (e.g., sedentary time, screen time, or both), the posture adopted (e.g., sitting, reclining), and any strategies they employed or considered to displace such behaviors (e.g., whether they are willing to displace their current sedentary or screen-based activity, cues that prompted movement, intended methods for breaking sedentary periods, or challenges faced in displacing sedentary activities) (Figure 1). Participants were asked to submit diary entries daily, with a minimum requirement of at least three entries per day for six out of the seven days to ensure the integrity of the research. Participants who did not meet the minimum requirement for diary entries received an email reminder. Participants were compensated between \$60 and \$75 based on their level of involvement; for those who completed all seven days of diary writing, we provided an extra \$15 as a bonus.

3.2.3 Debriefing Interview. We conducted a one-on-one debriefing interview of approximately 60 minutes after each participant completed the diary study. The introductory part of the semi-structured interview included an open-ended discussion, inviting participants to share their general thoughts about the diary study procedure and their perceptions of behavior displacement. We then discussed some specific entries that warranted further clarification through a shared-screen review of their submitted diaries on a Google Sheet. Before the debriefing interview, we summarized each participant's data, highlighting the total number of their responses regarding whether they displaced a certain activity or not. Participants were then asked about the challenges or barriers they faced when thinking about behavior displacement, as well as the opportunities they identified for potential displacement in their sedentary routines. Further inquiries explored participants' reflections on moments of challenge or reward in displacing sedentary activities.

3.3 Data Analysis

We processed the diary data by cleaning and categorizing the reported activities including the location, the activity purpose, the posture participants adopted, the fixture where the activity occurred, and the digital devices participants used. Specifically, activities were coded into categories such as work and studying, household, socializing, entertainment, religious or spiritual practices, and maintenance based on established frameworks [76], with reference to the ATUS Activity Coding Lexicons [82]. Some participants mentioned more than one activity per diary. In such cases, we coded only the activity that they considered to displace, if it was clarified in other diary entries or during the interview; otherwise, the first activity mentioned was coded. The lead author also conducted a codebook thematic analysis [8] on the diary's open-ended responses by first extracting the *contexts* of activities and participants' *rationale* for whether to adopt displacement and displacement strategies.

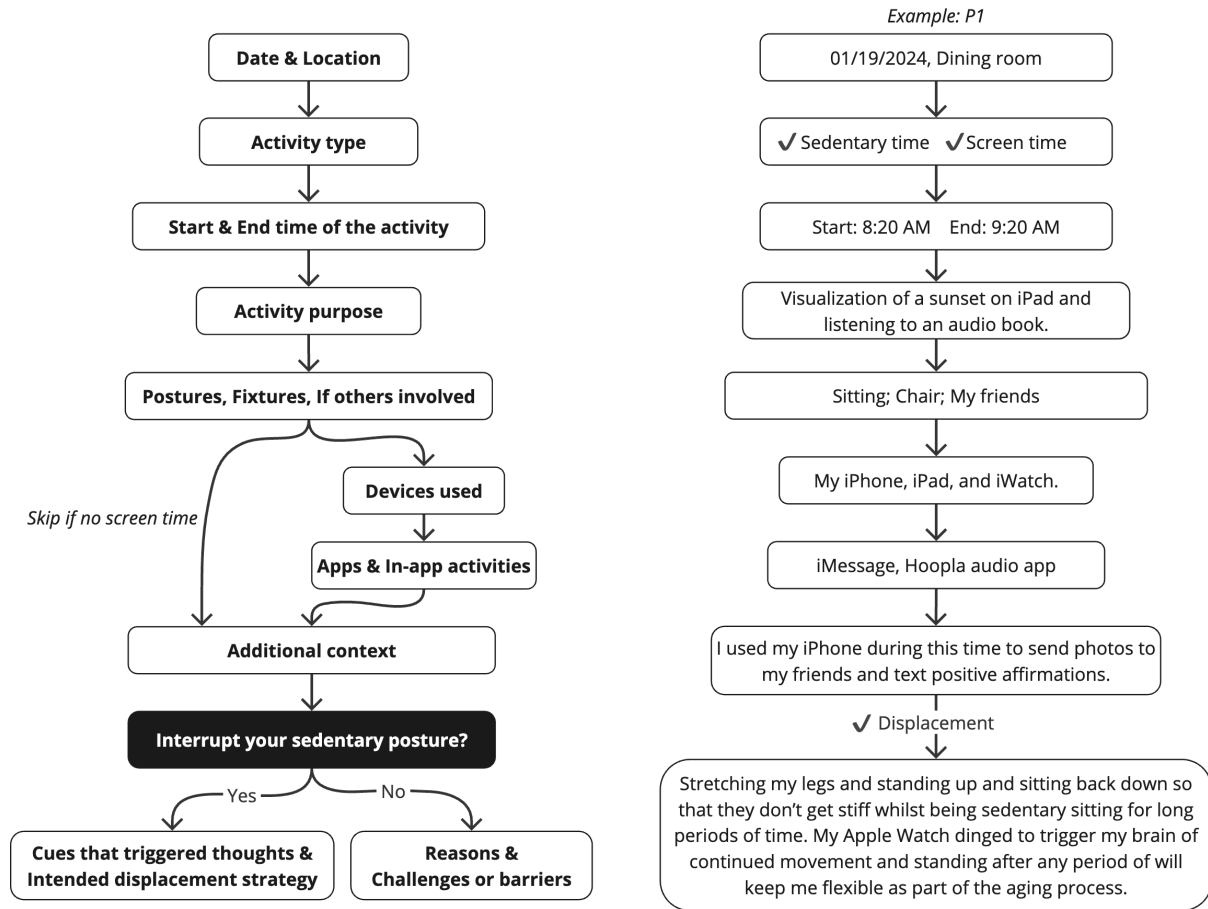


Fig. 1. A flow diagram of the diary design along with an example response from P1. For each sedentary or screen-based activity, we captured the date, location, activity type, start/end times, activity purpose, postures, fixtures, surrounding individuals, devices and apps used, and additional context. If participants considered the activity a good candidate for adopting displacement strategy, we asked for their reasoning (e.g., cues, intended displacement strategy); if not, we inquired about the reasons (e.g., challenges, barriers).

All debriefing interviews were transcribed manually with the assistance of Zoom Cloud transcription services. The lead author conducted a codebook thematic analysis [8] using the coding framework developed from the diary data analysis. The purpose of the debriefing interview analysis was to enrich the understanding of participants' decision-making processes regarding sedentary behavior displacement and the strategies they considered or implemented. Following Braun and Clark's guidelines [8], open coding was performed to expand and refine the initial coding framework. The research team, consisting of the lead author and advisor, reviewed and discussed the codes and potential themes in weekly meetings. The finalized themes, presented as follows, include: considerations on whether to displace; catalysts for when to displace; and strategies for how to displace sedentary behaviors.

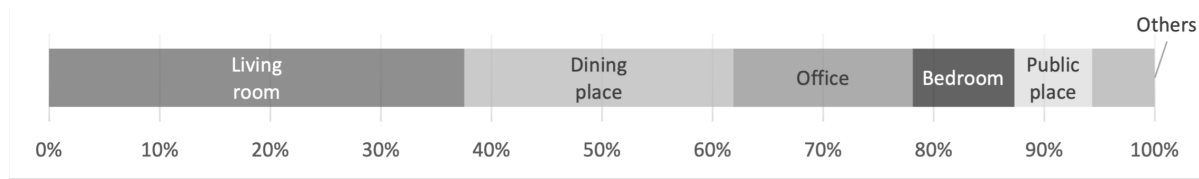


Fig. 2. Locations reported in the diary. Others include car or bus, porch, and bathroom.

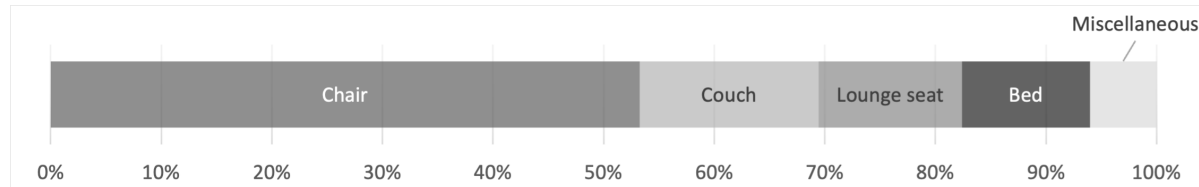


Fig. 3. Fixtures reported in the diary. Miscellaneous category includes transportation seat such as car seat and bus seat, bench, stool, and hot tub.

4 FINDINGS

In Section 4.1, we first present descriptive statistics of the structured diary data. In Section 4.2, we examine the key contexts influencing older adults' decisions to displace sedentary behaviors, exploring their considerations involved in balancing physical movements with cognitive, emotional, and social engagement. Following this, we detail two behavior displacement strategies participants employed—sequential displacement in Section 4.3 and concurrent displacement in Section 4.4—along with the contexts in which these strategies were used. Our findings offer a deeper understanding of how older adults incorporate displacement into their daily life to support both physical well-being and activity engagement.

4.1 Descriptive Statistics of the Diary Data

Participants submitted 332 diaries in total, in which there were 223 (67.2%) responses of “Yes” and 109 (32.8%) responses of “No”, to the question of whether they are willing to displace their current sedentary or screen-based activity. Among all the “Yes” responses, 24 (10.8%) diaries reflected conditions where participants considered displacement as a possibility but did not implement it, while 199 (89.2%) documented cases where participants really implemented displacement strategies into their sedentary time. Participants submitted 232 (69.9%) diaries involving both sedentary and screen-based activities, 80 (24.1%) diaries about sedentary activities, and 20 (6.0%) about screen-based activities.

The reported locations of sedentary or screen-based activity included living room (37.6%), dining place (24.3%), office (16.2%), bedroom (9.2%), public places such as restaurant and hospital (7.0%), car or bus (4.1%), porch (1.4%), and bathroom (0.3%) (Figure 2). Most activities were performed while sitting (95.8%), with a smaller proportion while lying down (4.2%). Participants reported the fixtures where activities occurred, which involved chair (53.2%), couch (16.2%), lounge seats such as recliner and rocker (13.0%), bed (11.6%), transportation seats such as car seats and bus seats (4.0%), bench (1.0%), stool (0.5%), and hot tub (0.5%) (Figure 3). Participants engaged in more than half of the reported sedentary activities alone (51.3%), while others involved family members (28.0%), friends (8.1%), and others (12.6%) such as doctors and repairmen (Figure 4). The reported screen-based activities predominantly involved smartphone (34.5%) and laptop (32.9%) devices, with additional devices including TVs (17.8%), tablets (9.3%), PCs (4.7%), and smartwatches (0.8%) (Figure 5).

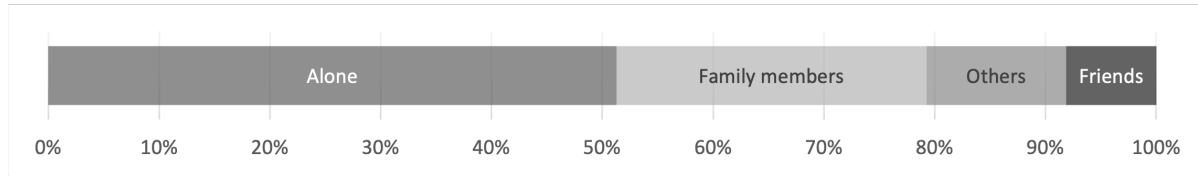


Fig. 4. People involved reported in the diary.

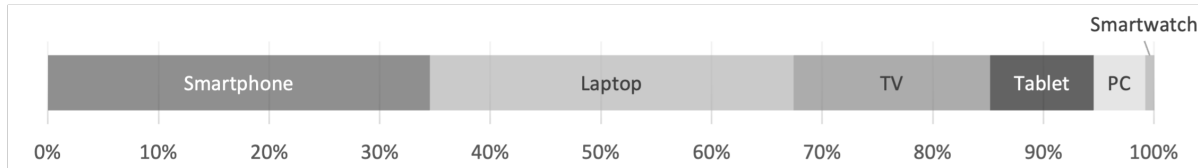


Fig. 5. Devices used reported in the diary.

4.2 Considerations on Whether to Displace Sedentary Behaviors

We explored the considerations participants made in deciding whether to displace sedentary behaviors, focusing on the conditions that shaped their willingness to displace. Participants' reflections revealed both motivations that encouraged displacement and constraints that hindered displacement.

4.2.1 Motivations to Displace. Older adults' decisions to displace sedentary behaviors are often guided by internal and external motivators, including physical discomfort, health awareness, and routine responsibilities.

Mitigate Pain and Discomfort. A common motivation for displacement was the desire to *relieve physical pain or prevent discomfort*. Participants reported displacement in response to stiffness from prolonged sitting (P1, P5, P7, P9, P11) or cold temperatures during the study period (P5). For example, P1 reported “*getting stiff after sitting for 2 hours*” and P6 noted “*lower back pain*” as triggers, while P7 wrote that “*my back hurting was the cue to stand up and move back and forth*.”

Concerns about the health consequences of prolonged sedentary behaviors also emerged as a strong motivator (P2, P3, P9). P9 reflected that “*the cues were thinking of these big blocks of sedentariness I am now realizing I have*,” highlighting an awareness of sedentary patterns, which then motivated displacement. P1 shared a similar proactive attitude toward staying active, “*standing after any period of [sitting] will keep me flexible as part of the aging process*” and “*staying in movement is vital as I age so I can enjoy retirement fully*,” indicating an awareness of the long-term benefits of regular movement.

Responsibilities that Prompt Movements. Routine responsibilities such as daily chores and household management tasks often motivated participants to displace their sedentary behaviors (P4, P5, P6, P7, P11, P13). For example, P5 stated that she decided to complete the laundry to break her screen time: “*The buzzer on the clothes dryer dinged and when I heard it, I chose to leave the computer to take garments out*.” P6 also resonated with this, “*I knew that I had this time free and last evening I became aware that my supplies of yarn were in a mess and needed to be cleaned up*,” so she decided to “*tackle this chore in the amount of time before starting checking emails*.” P11 described the high volume of household responsibilities as a motivator for movement, “*Household chores are extensive right now... so the sit time probably would have been longer if I didn't 'have to' get up to tackle chores and errands that must be done*.” This suggests brief but frequent opportunities for displacement embedded in older adults' daily life.

Pets were another motivator for displacement within participants' routines. As P9 noted, *"took short break (between piano practice sessions) to pet the cat,"* illustrating how even brief interactions with pets can encourage breaks in sedentary time.

4.2.2 Reasons to Remain Sedentary. While many participants were motivated to displace sedentary behaviors, they also described a range of reasons why they preferred not to displace. These include attention demands, perceived impacts on productivity and quality, physical fatigue, and social norms.

Attention Demands in Activities. Participants' decisions not to displace sedentary behaviors were often shaped by the level of attention required for the task at hand. Many participants explained in the interview that displacement would be feasible when it would not interrupt their focus. As P7 stated, *"the only issue is that you have to be at a point where you can make a change without interrupting whatever you're doing at the moment."* While some activities allow for divided attention, tasks requiring deep thought, concentration, or emotional engagement often hinder displacement. These activities include work-related duties (P2, P7, P9, P11), social interactions (P2, P10, P11), and religious and spiritual practices (P1, P2).

Work contexts were particularly demanding. For example, P9, a psychotherapist, shared that *"Doing psychotherapy takes my full concentration, it would be disruptive if I stood up."* P2 and P7 described needing sustained attention for client interactions or form completion, where even brief movement would disrupt their workflow.

Social interactions sometimes constrained displacement. P2 described being too focused on the conversation with her daughter to consider moving, while P11 noted distractions from multitasking during socializing: *"I feel distracted when I do this—neither paying attention to the person on the phone, nor the material I'm reading."* Still, P11 acknowledged some social conditions allowed displacement: *"I think some calls require or deserve undivided attention, but it was easy and satisfying to talk and do gentle qigong while talking and not feel distracted."*

Mindfulness activities also required a high degree of focus. P1 described her meditation as a time demanding concentration, where she focused on reprogramming her brain using DBT (Dialectical Behavior Therapy) techniques to change longstanding unhealthy thought patterns stemming from childhood trauma, *"other activities would be a distraction to me internalizing what I need to change."* However, she added that as she became more skilled, she might integrate displacement: *"I think in the future I will be able to add movement once I learn how to actively listen,"* which corresponds to the reduced attention required as the activity became familiar.

Perceived Impacts on Productivity and Quality of Activities. Many participants chose to remain sedentary not only for work-related reasons, but also due to immersion in leisure activities or the desire to complete tasks efficiently.

Entertainment immersion was frequently mentioned by participants as a reason not to displace, including watching movies or TV (P3, P4, P5, P9, P13), playing the piano (P9), reading (P9), and relaxing with a cigar (P10). For example, P3 experienced being attracted by a movie, *"the movie was good so I did not want to get up,"* while P9 described an instance when watching TV: *"I could have interrupted the activity, but I forgot... I was perfectly comfortable watching the show which was funny and engrossing."* Similarly, P13 shared a game watching scenario, *"I usually stay with a game during the first half, especially one that is social as this one was."*

Personal interests also played a role in staying sedentary. P4 enjoyed watching cooking videos, while P9 described playing the piano as *"very absorbing"* and reading articles as both *"restless"* and *"very absorbed."* P10 described his choice to remain seated and enjoy a cigar, *"Movement would have disrupted my peaceful time."*

Several participants emphasized the importance of *completing tasks in one sitting*, particularly when under time pressure (P2, P3, P4, P5, P7, P11, P13). P7 described being *"disorganized and hyper-focused"* due to last-minute paperwork, while P11 aimed to complete short projects in a single session, *"When I work on a short project like this, I try to get it done in one seating."* P13 echoed this to *"get through this task quickly so I could get back to my daily chores,"* choosing to keep her productivity without displacement.

Physical Fatigue. Fatigue from daily activities or prior exercise was a common reason participants chose not to displace sedentary behaviors. While physical activity was valued, the need for rest and recovery was equally important. For example, P2 shared, *“Although I had thought of getting up and moving, I was tired from the week and needed to devote my energy to reading and reflecting thoughtfully.”* Similarly, P3 described her relaxing morning as a welcome break: *“I am tired since I woke up earlier than usual today so it was nice to just sit and watch Hallmark movies, the news, and late-night shows.”*

Some participants “banked” earlier physical activity to justify later sedentary time. After a “high-intensity workout class”, P7 explained, *“I was tired and felt I banked enough time with the class to make up for the sitting during work.”* P10 echoed this logic, *“I just completed 90 minutes of biking and I was a bit sore... I felt I had gotten enough exercise for the day,”* highlighting how recent exertion influenced participants’ willingness to displace.

Social Norms. Social contexts also influenced decisions not to displace. In formal settings like lectures or work meetings, participants avoided movement to prevent disruption. As P1 noted of a community lecture, *“Lecture did not include anything except listening and speaking, any kind of movement would be disruptive for the speaker.”* The constraint extends to informal settings as well. P3 felt restricted by the possibility of needing to respond during a Zoom meeting, *“I felt that I could not get up from my computer in case someone came out of the room with a question.”* Similarly, P7 and P8 described moments where social or professional obligations prevented movement, as P7 recalled, *“I was behind on my work and had a patient contact me right as I was going to stand up. Had to sit for another hour to take care of him.”*

4.3 Sequential Displacement

In conditions where participants considered it appropriate to displace their sedentary behaviors, they frequently chose displacement as a conscious strategy to enhance productivity, physical well-being, or mitigate discomfort. Following Rhodes et al. [71], we categorized these strategies into sequential and concurrent displacement (Figure 6), based on whether the goals at hand are competing or congruent. We further detailed subtypes within each category and proposed a comprehensive framework for behavior displacement strategies.

Sequential displacement involves replacing one sedentary behavior with a different activity either immediately or after some time. It often occurs when the goal of the sedentary activity (primary goal) and the goal of the desired physical activity (secondary goal) are competing.

4.3.1 Take a Break Within a Sedentary Activity. Taking short breaks for standing, stretching, or movement was a common sequential displacement strategy among participants, especially to alleviate discomfort and stiffness from prolonged sitting (Figure 6 (a)(1)). For example, P1 described responding to discomfort by incorporating stretching into her routine, *“My back started hurting, and so I stood and stretched.”* Over time, this became habitual: *“Standing up to trigger me out of a sedentary position and into rolling my neck around and shrugging my shoulders to relax what’s tense above my neck.”* P5 also integrated standing balance exercises: *“Sitting to standing, first on both legs and then on one plus sit to stand... Got up and did a series of balance exercises for 15 minutes.”* Others adopted simple routines like walking for water every 30 minutes (P10) or stretching on the couch (P11).

Participants also found natural opportunities to displace during screen-based activities. P5 took a break during TV to get new yarn and walk the hallway. P13 used pauses in movies or TV shows to stretch and move around, while P10 noted, *“The cues were the commercial breaks. I should have exercised/lifted weights during the show.”*

For many, these breaks were not interruptions but opportunities to refresh both physically and mentally. P2 described walking and stretching during a professional development meeting to *“have a mental as well as a physical break,”* and also paused between emails to do sit-ups. P3 used breaks to relieve from unenjoyable work: *“Grading papers is one thing I do not enjoy about the consulting work that I do. I broke this task up a bit by doing PT band exercises in my chair every three papers.”* P8 shared a similar feeling that displacement improved his

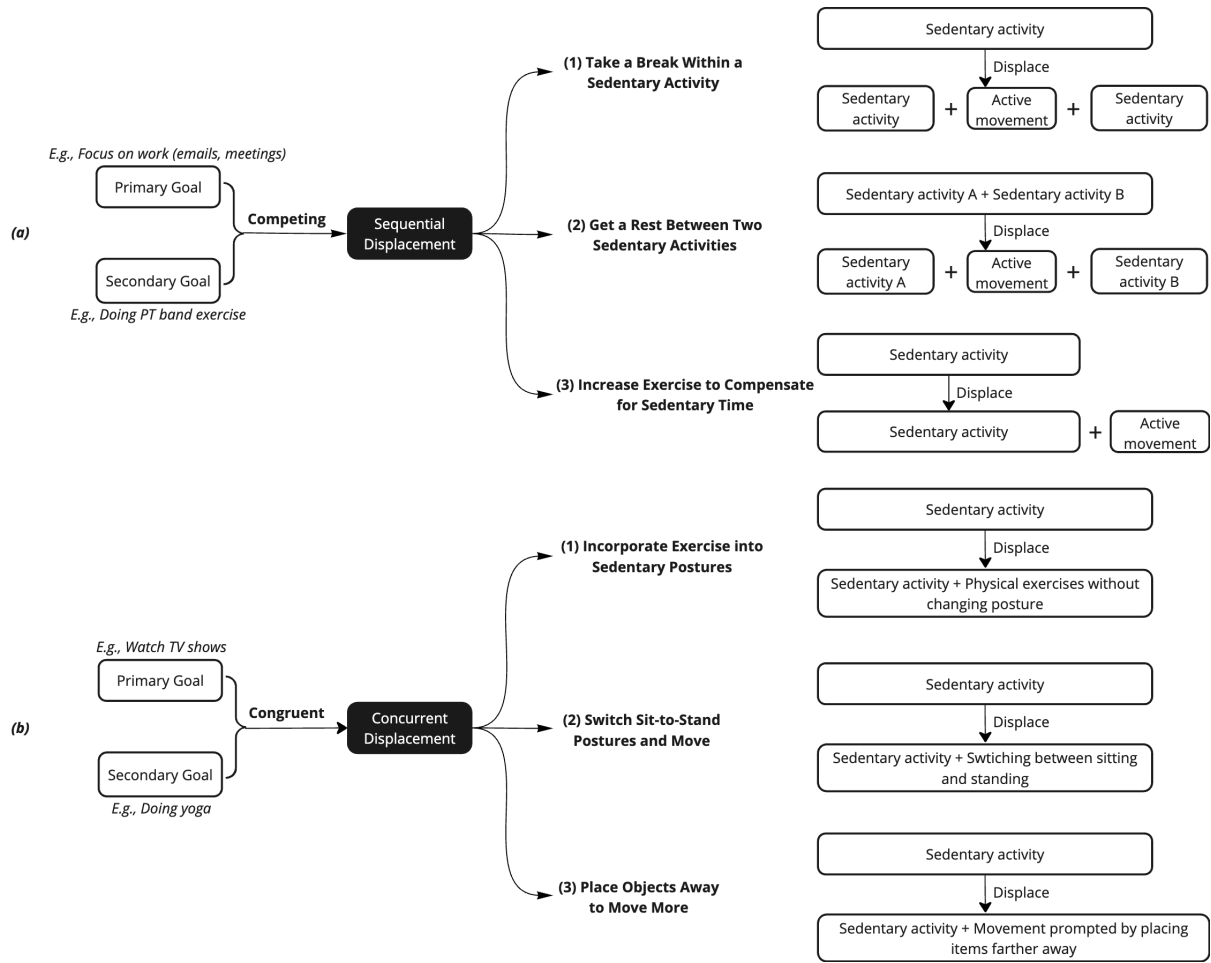


Fig. 6. Two types of displacement strategies identified in the study: (a) Sequential Displacement and (b) Concurrent Displacement. Sequential displacement involves replacing one sedentary behaviors with another activity, typically when the goals of the two activities compete. Examples include: (1) taking breaks within a sedentary task, (2) moving between two sedentary activities, and (3) increasing exercise to compensate for sedentary time. Concurrent displacement refers to integrating physical activity within an ongoing sedentary task without interrupting it, usually when the goals of the activities are congruent. Examples include: (1) incorporating exercise into sedentary postures, (2) switching between sitting and standing, and (3) placing objects at a distance to encourage movement. The diagram represents how each strategy displaces sedentary behaviors by adding active movement, either in sequence or simultaneously with the ongoing activity.

well-being while working: “I feel better than I have in a long time by including exercise and movement during my sedentary time.”

Some participants emphasized the value of intentional and meditative movement during breaks. P11 chose tai chi over household tasks:

During the "get out of the chair" time, instead of doing things like dishes, laundry, etc., did more walking tai chi and active movement only - swinging arms, wall planks, etc. so that the break movement were focused on the movement and meditative thoughts - not just on my feet getting things done.

Some participants also viewed displacement as a chance to reinforce cognitive tasks. P1 described combining light chores with learning:

I would like to help with my short term memory whilst including mopping and sweeping my apt. helping with not being sedentary whilst learning new information. Which could be a reinforcement for learning a new language improving my memory.

4.3.2 Get a Rest Between Two Sedentary Activities. Many participants intentionally inserted movement between two sedentary activities as a form of displacement (P2, P5, P7, P13) (Figure 6 (a)(2)). P2 illustrated this by pausing before beginning a new activity: "After my previous sedentary activity, I purposely walked around and changed my clothes before writing, rather than writing the letter at the same time." These transitions often involved light chores, as when P7 noted, "I volunteered to walk to the grocery store and carry bags back, cook the dinner and clean—all done on my feet with a little dancing shuffle in between."

Others described brief transitions between hobbies. P5 used the end of a game as a cue to move: "At the end of the game I got up and did a series of balancing postures and also walked throughout the house as I had completed my knitting project and needed to get new yarn for the next one." Similarly, P13 broke up leisure time to reduce prolonged sitting: "I decided to break up the two activities and got up to sort laundry before starting Sudoku."

4.3.3 Increase Exercise to Compensate for Sedentary Time. Recognizing the potential health impacts of prolonged sedentary behaviors, participants identified increasing their physical exercise as a strategic displacement (P4, P5, P11) (Figure 6 (a)(3)). For example, P4 chose yoga to compensate for extended periods of sitting, especially after long drives: "Yoga at home after long-time sitting in car." P5 integrated physical activity into daily chores, considering them as opportunities for movement: "I could tackle this chore (prepare donations for Good Will delivery) in the amount of time before starting lunch." Similarly, P11 described a movement plan to counteract a morning spent mostly in sedentary activities: "Will now do some stretches, tackle chores that require being on my feet (with music) and then go to weekly farm market for this week's produce. Hopefully go to the pool again this afternoon...all to offset the morning's several hours of pretty much sitting around."

4.4 Concurrent Displacement

Concurrent displacement, on the other hand, refers to integrating a different behavior within an ongoing sedentary activity without interrupting it. This strategy is typically used when the goals are congruent, enabling participants to maintain engagement in the sedentary activity (primary goal) while simultaneously fulfilling their physical activity goals (secondary goal).

4.4.1 Incorporate Exercise into Sedentary Postures. Participants found various ways to incorporate physical activity while remaining seated or reclined, from using weights and resistance bands to modifying posture and creating space for movement (Figure 6 (b)(1)). These strategies allowed movement without interrupting ongoing sedentary tasks. For example, P1 described integrating weights into her mental exercise routine, "I included weights to today's (DBT) exercise," and expressed a broader goal of aligning movement with cognitive practices: "Reprogramming my brain to match exercise and DBT practice skills – build tolerance to exercise at the same time as practicing my DBT skills." P10 also planned to incorporate weight training during screen time: "I intend to have hand weights next to the couch so I can do curls during the shows."

Participants also used seated time for stretching or resistance exercises. P1 added pilate instruments for arm stretches and leg squeezes while sitting erect, whereas P3 used a PT band for shoulder exercises on the sofa. P9 practiced balance and knee exercises while watching TV and stretched her legs during professional calls: "I move

in my chair though and sometimes stretch legs out of view of the client." P11 described waving arms and shaking legs as a "fan" action to keep moving and noted the importance of choosing seats that allow space: *"Sitting in my chair is a better place for seated movement; the couch isn't in the way to swing arms, kick legs, do side bends."*

Some participants also focused on posture as a subtle but intentional strategy. P2 made a point to sit upright while typing, and P3 preferred sitting in a structured chair rather than reclining. P11 commented on the benefits of sitting at the table: *"My posture is better at the table,"* and considered adjusting her car seat to allow for light stretching while seated.

Beyond physical activity, this strategy also supported emotional and mental well-being. P1 noted the benefits of incorporating exercise into meditation, *"I feel better with movement when I feel depression or anxiety coming on; it really helps my anxiety subside."* P8 shared a similar sentiment about the therapeutic value of movement during sedentary time.

4.4.2 Switch Sit-to-Stand Postures and Move. Adopting sit-to-stand postures was another key strategy for participants to displace their sedentary behaviors without interrupting the ongoing task, particularly when the activity allowed for divided attention (Figure 6 (b)(2)). Participants found natural breaks or lighter moments to stand, stretch, or walk short distances. Several participants implemented this strategy during online sessions. P2 described that familiarity with webinar content enabled her to displace without sacrificing engagement: *"I did know some of the information which was discussed, so I was able to get up from my chair and walk a short distance while listening to the webinar."* Similarly, during an online prayer session, she tried *"walking around while he spoke before beginning the prayers,"* explaining that routine participation made her feel comfortable doing so.

P7 echoed this strategy, describing his behavior during a Zoom session,

The Zoom session required concentration, however the beginning of the conference recording and several emails didn't require as much focused attention, so I could stand up, stretch and bend, so walk a short distance and back while still listening to the recording.

This strategy also extended to leisure and household activities. P5 switched from sitting to standing while playing games and watching TV. P11 described engaging in gentle qigong while talking on the phone, reflecting, *"It was easy and satisfying to talk and do gentle qigong while talking and not feel distracted."* P1 similarly found movement during video calls with family to be uplifting, sharing that stretching while FaceTiming made her feel *"more connected in body and spirit."* P3 adopted a routine of doing PT exercises while waiting for coffee to brew or soup to heat: *"I have started doing PT exercises while my coffee is brewing or soup is being microwaved."*

In some cases, this strategy included more intensive movement. P12 displaced his TV time with indoor physical activity: *"I walked quickly for an hour on treadmill and then transferred to cross-trainer while watching TV."*

4.4.3 Place Objects Away to Move More. Another innovative strategy employed by participants to displace their sedentary activities was to strategically place objects further away from their spot, thereby necessitating movement to accomplish tasks (Figure 6 (b)(3)). For example, P2 leveraged meal times as opportunities for displacement by deliberately putting dishes and foods at a distance from her seat: *"Purposely leave different foods and dishes away from my chair so that I needed to get up and walk to get them."* This approach was also extended to fetching additional servings with movement between the dining room and kitchen, *"I purposely went back and forth to get additional food."*

This strategy was not only limited to the home settings. P4 adopted this method in the context of traveling and parking, choosing to park her car farther from the destination than usual: *"Parked farther away for a longer walk to and from the dentist's office."* Similarly, P5 found value in this practice, *"It felt good to have parked a bit away from the front door of the building where we met."*

5 DISCUSSION

In this section, we discuss design implications for adaptive interventions that promote behavior displacement among older adults. First, we outline approaches to timing interventions and integrating movement into routines to minimize disruption. Next, we highlight the role of motivational alignment and personalization to address individual needs and preferences. Throughout, we emphasize user agency in designing effective interventions. Finally, we reflect on methodological insights and limitations, identifying opportunities for future research to expand the scope of our work.

5.1 Leveraging Low-attention Moments for Promoting Behavior Displacement

Our findings reveal that older adults often consider attention demands and their desire to maintain activity productivity and quality when deciding whether to displace sedentary behaviors. This decision-making process highlights the importance of designing adaptive systems that align with attention management. In particular, principles from interruptibility models demonstrate that digital interventions should adaptively prompt movements at moments of low attention demand, minimizing disruption and aligning with users' ongoing activities [59]. Prior work on workplace engagement by Mark et al. [49] observed that individuals exhibit varied patterns of focus and engagement throughout the day, influenced by the nature of their tasks and time of day. This rhythm of engagement suggests that interventions for behavior displacement could be customized to align with individuals' natural fluctuations in attention, effectively enhancing both physical and mental well-being.

For activities that can be handled with divided attention, participants in our study reported more freedom to introduce displacement without interrupting the activity's integrity. For example, P2 and P7 shared that during webinars or Zoom sessions, they often stood up, stretched, or walked indoors while still listening. Similarly, during leisurely activities like watching TV or reading, participants (P2, P9, P10, P11) incorporated light movements—like leg stretches or arm exercises—leveraging these low-attention moments without sacrificing engagement quality. Conversely, activities necessitating full attention—such as work-related tasks, social interactions, and engagement in puzzles or games—posed challenges for displacement. Participants expressed concerns that introducing displacement could disrupt their focus. As such, intervention systems should recognize these high-attention periods and avoid prompting movement that could hinder productivity or quality of sedentary activity.

Adaptive systems like those described by Zhang and Rehg [91], which dynamically modeled attention patterns based on contextual data, and Srivastava et al. [77], who demonstrated the utility of gaze features for identifying attention-demanding tasks, further highlight the potential to identify “low-attention” moments as opportunities to encourage movement. For example, wearable devices or smart home technologies could monitor signs of fatigue or prolonged sitting, delivering context-aware prompts like gentle vibrations or ambient lighting changes to propose displacement that aligns with older adults' current state.

5.2 Designing Routine-respecting Interventions

One significant finding from our study centers on participants' preference for *integrating* movement into existing routines rather than *substituting* sedentary behaviors entirely. While previous interventions have focused on reducing overall sedentary time and increasing physical activity level through substitution [6, 73], our participants emphasized displacement strategies that complement rather than disrupt their routines. This aligns with their inclination toward low-exertion activities, such as stretching during TV watching or indoor ambulations while on phone calls, which echoed with previous studies on small movements and context-aware activity integration [29, 33, 87]. Such preferences suggest that adaptive systems should prioritize integrating movement into sedentary routines, minimizing cognitive and physical effort.

This approach mirrors insights from Luo et al.'s break-prompting system (“Time for Break”), which enabled information workers to incorporate standing or stretching into their workflow [44]. For our participants, whose

sedentary behaviors are often connected with screen time, context-aware prompts (e.g., reminders during video pauses or app transitions) could leverage device-level awareness to reduce resistance to behavior change. Participants frequently relied on digital calendars and emails to manage tasks, suggesting opportunities to embed movement prompts within these tools. Contextual data such as calendar events and app usage can reflect nuances in activity types, for example, distinguishing between professional meetings, leisure activities, and periods of focused work. Prior studies have similarly demonstrated that integrating behavioral context (e.g., daily schedules, app interactions) into intervention timing can enhance relevance and minimize disruption [32, 58]. Drawing on such contextual information can help systems avoid treating all prolonged sitting uniformly and better align prompts with moments where breaks are beneficial for both physical and cognitive well-being. Tools like Digital Avatars [7], which monitor activity patterns and provide reminders for scheduled tasks, further showed how systems can integrate displacement strategies into daily routines while supporting proactive health management.

By aligning with—rather than competing against—existing routines among older adults, such systems exemplify the potential for routine-respecting interventions without sacrificing productivity and quality of sedentary time. Butte and Levy’s findings further support this approach, suggesting that even brief and context-aware transitions can also improve physical function measurably [9]. For older adults, linking interventions to familiar tools and workflows could enhance receptivity, as systems adapt to their routines rather than imposing external demands. This strategy also respects their agency while promoting behavior change through existing habits.

5.3 Toward Context-aware JITAI Systems

Our findings suggest rich opportunities for developing more context-aware, personalized just-in-time adaptive interventions (JITAI) to support behavior displacement among older adults. JITAI aim to deliver timely support that adapts dynamically to individuals’ changing contexts, providing assistance when it is most needed and likely to be effective [56]. While current wearable and mHealth applications such as Apple Watch and Fitbit already issue generic movement prompts based on tracking data [13, 85], our findings highlight the need for JITAI that are designed to the nuanced decision-making processes of older adults.

Older adults in our study often mentioned physical discomfort and stiffness as immediate internal cues for displacement, pointing to a clear opportunity for systems to detect signs of prolonged sitting from wearable devices or heart rate variability and deliver timely, low-effort movement suggestions. Prior work on sedentary interventions has primarily triggered prompts based on fixed sitting durations [54], yet our findings suggest that simply detecting time thresholds is insufficient. Participants’ willingness to displace sedentary behaviors was deeply connected with their attentional states and their engagement with ongoing activities. Activities that demanded full cognitive or social engagement, such as psychotherapy sessions or spiritual practices, were often preserved without interruption, reflecting concerns about maintaining productivity, quality, and social appropriateness. These findings echo the broader literature on receptivity-aware JITAI design, which emphasizes that effective intervention timing must account for users’ cognitive availability and willingness to act [14].

In this context, customized variables for JITAI should go beyond physical indicators like accumulated sedentary time to incorporate contextual cues such as attention demands, activity types, social settings, and locations. For example, mobile application usage patterns—such as frequent app switching or entertainment app use—have been associated with lower attention focus and higher receptivity to interruptions [11]. Integrating these behavioral indicators into JITAI could help systems infer when older adults are more receptive to movement cues without disruption. Moreover, recent work on microinteraction EMA (μ EMA) suggests that even lightweight prompts can suffer from contextual non-response biases depending on time of day, device usage, and user activity [67], further demonstrating the need for adaptive, context-aware delivery strategies.

Our participants’ use of both sequential and concurrent displacement strategies also offers insights for designing flexible JITAI content. When older adults experienced natural transitions between tasks or brief cognitive pauses,

they often engaged in sequential displacement, choosing to take breaks or insert physical activity between sedentary sessions. In contrast, during prolonged engagements where attention could be divided—such as casual phone calls or TV watching—participants adopted concurrent displacement strategies, integrating subtle movements without abandoning the primary activity. Future JITAIs could leverage such patterns by offering different types of movement suggestions depending on the inferred nature of the user’s engagement: encouraging brief, restorative breaks after completing tasks, or suggesting stretching or posture adjustments during ongoing, less demanding activities. Incorporating historical responsiveness patterns such as preferred types of displacement, reaction times, and ignored prompts, can further adjust to individual evolving needs.

5.4 Personalizing Interventions for Individual Variability

Adaptive systems should account for the heterogeneity of older adults’ lifestyles, preferences, and life circumstances to effectively support behavior displacement [27]. Our participants showed diverse decision-making patterns for displacing their sedentary time. For example, while some participants viewed sedentary activities like watching TV and reading as relaxation time that cannot be interrupted, others incorporated stretching or ambulation into these moments. Such variability stresses the need to move beyond one-size-fits-all approaches.

Differences in life circumstances among older adults—such as employment status (working, retired) or caregiving responsibilities—further shape their daily routines and displacement opportunities. For example, participants like P7 and P8 balancing part-time work would prioritize movement during screen-based tasks, whereas retirees like P1 and P11 would prefer integrating displacement into leisure activities. Systems could leverage reinforcement learning frameworks, such as HeartSteps [43], to dynamically adapt prompts based on user feedback and historical behavior. Caldeira and Chen [10] emphasized that older adults’ engagement with self-tracking tools often prioritizes health management over explicit behavior change, guided by practical needs and preferences. This observation highlights the importance of designing systems with flexible tracking and low-barrier interactions, such as step monitoring or light exercise suggestions based on contextual cues. By aligning interventions with personal values and goals, systems can enhance sustained engagement in older populations.

5.5 Supporting Agency in Displacement

Agency in behavior displacement among older adults is an important insight in our study, where participants consistently expressed a preference for maintaining control over how and when they employ displacement in their sedentary routines. Agency refers to individuals’ ability to act independently and make their own choices, which in older adults pertains to managing sedentary behaviors and integrating physical activity into their daily routines [6]. Recent work on user-centered JITAI design similarly emphasizes that interventions should preserve users’ autonomy by offering flexible, contextually relevant options rather than rigid suggestions [36]. Building on this perspective, along with Kim’s exploration of autonomy, particularly the dimensions of capacity for action and self-governance [37], we advocate for designing systems that empower users to make autonomous decisions about displacement, adapting to their routines, preferences, and momentary contexts.

The importance of autonomy in behavior change is well-documented in self-determination theory [20], which highlights the role of intrinsic motivation in sustaining long-term engagement. Building on this foundation, HCI research has demonstrated that user-centered design principles—emphasizing choice, personalization, and respect for users’ values—can significantly enhance the engagement with technological interactions [21, 81]. For behavior displacement, this means designing systems that offer users a range of options and allow them to initiate or modify prompts in ways that feel empowering. For example, our participants emphasized the need to allow them to adjust the timing, intensity, and type of physical activity to suit their individual preferences and circumstances when considering displacement strategies. Older adults expressed a desire for flexibility in choosing between low-exertion activities and more vigorous physical activities based on their energy levels or

immediate priorities. Wang et al. demonstrated that older adults identified multiple values guiding their activity engagement, such as prioritizing feasible and minimally intrusive activities [87]. Designing systems that provide a customizable menu of movement suggestions could enable users to select activities that align with their mood, physical condition, or time constraints. By emphasizing activities that integrate into existing routines, systems can enhance users' sense of agency while fostering sustainable displacement behaviors.

Inclusive design is another critical consideration in supporting agency and flexibility for older adults. Systems should accommodate diverse physical abilities and emotional needs [88], ensuring that all users can engage with displacement effectively. For example, our participants emphasized the value of sedentary relaxation when feeling tired and keeping sedentary hobbies such as online reading or digital games. While many existing interventions target general populations or clinical settings, fewer designs explicitly account for the preferences and constraints of aging populations [4, 25]. Our findings highlight the potential of user-centered and inclusive approaches to empower older adults, fostering a sense of ownership over their health and well-being. Future research should explore how these principles can be further refined, such as integrating real-time feedback on user preferences or studying the long-term impact of agency-driven interventions on behavior change.

5.6 Limitations and Future Work

The demographic composition of the study sample might not accurately represent the broader older adult population. Most participants were highly educated and had high-baseline technical proficiency to use Zoom. The age range of participants (65–83 years, $M = 71.8$) excluded the “oldest old” (85+), who often experience greater physical and cognitive challenges and higher sedentary levels [85]. While most participants reported a few health issues such as arthritis or diabetes, none had severe disabilities or disorders that would significantly impact mobility or sedentary routines, consistent with the study exclusion criteria. Although the findings showed that living situations such as living alone, with spouses, or in an intergenerational family, can influence participants' displacement decisions, we did not collect data on household setting or income status, and participants were mainly residents of urban communities. Moreover, the diary study method may have introduced the possibility of reactivity effect [57]. The act of jotting down these activities and thoughts likely increased participants' awareness and increase displacement activity. The one-week study duration may also have been insufficient to capture recurring or seasonal patterns in sedentary behavior and displacement strategies. Longer study periods could provide more comprehensive insights into habitual behaviors. Seasonal timing during the winter months may have further influenced displacement, as cold weather likely increased indoor sedentary time and limited outdoor activity. Our study offers behavioral insights and design implications that can inform future work. An important next step is to develop mobile and wearable interventions that build on these findings. We see promising opportunities to explore adaptive algorithms that support real-world displacement strategies, designed in alignment with the routines, preferences, and agency of older adults.

6 CONCLUSION

In this study, we conducted a 7-day diary study and debriefing interviews to investigate how older adults decide whether and how to displace sedentary behaviors. Our findings reveal that decisions to engage in displacement are influenced by factors such as attention demand, productivity, physical fatigue, and social norms. We categorized the sequential and concurrent displacement strategies that older adults employed in various contexts. Older adults actively manage their sedentary time by weighing the benefits of displacement and engagement of sedentary activity, demonstrating agency in maintaining their quality of life. We highlight the importance of personalized, adaptive interventions that integrate movement into daily routines, respecting the diverse preferences and behavioral goals of older adults. Our work contributes to the understanding of behavior displacement and offers insights for designing interventions that promote displacement among older adults.

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